SOC Incident Report

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Alert Source: Windows Event Security Log (CSV)
SIEM: Splunk
Lab: BTLO Windows Events

1. Incident Title

Windows Brute-Force Login Attempt Detected

2. Executive Summary

A brute-force login attack targeting the administrator account was detected in windows security logs. Multiple audit failure events occur over a short time window. Using Splunk, missing fields were created using Field extraction(Regex) to normalize the log for analysis. SPL queries were used to determine the targeted account, source IP, port range, number of failures, and the timeframe of the events. Threat intelligence confirmed the source IP as an external confirming a high-severity attack.

3. Log Preparation / Field Extraction

- The CSV logs were missing some important fields: Source IP, Source Port, Account Name.
- Field extraction(Regex) was used to extract the missing fields
- This will ensure SPL queries return accurate results for the investigation.

4. Investigation Findings

SPL Queries and Analysis

	Queries	Findings	Purpose
Count of Audit Failure	index=main sourcetype="csv" "Event ID"=4625 TargetUserName="administrator" |stats count as "Total Audit Failure"	3103	Determines the total number of failed login attempts targeting administrative account

Failure Reasons *See figure 2*	index=main sourcetype="csv" “Event ID”=4625 TargetUsername="administrator" | stats count by Failure_Reason	Unknown username or bad password	To determine the reasons for the failed login attempts
Source IP address	index=main sourcetype="csv" “Event ID”=4625 TargetUsername="administrator" | stats count by SourceIP	Attack originated from 113.161.192.227	To Identify attacker’s location
Port Range	index=main sourcetype="csv" “Event ID”=4625 TargetUsername="administrator" | stats min(SourcePort) max(SourcePort)	49162 - 65534	Determines Automation Behaviour
Timeline of Failed Attempts *See figure 1*	index=main sourcetype="csv" “Event ID”=4625 TargetUsername="administrator" | timechart count span=5m	Attack occurred in bursts over the observed time frame	Visualize attack patterns

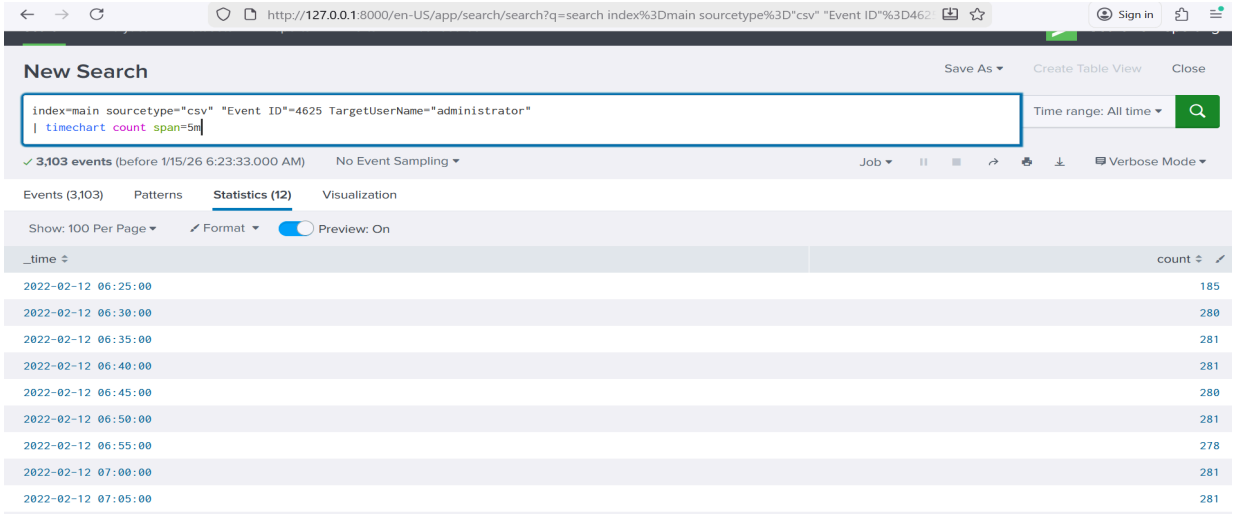

Figure 1: SPL queries and result for timeline of failed attempt.

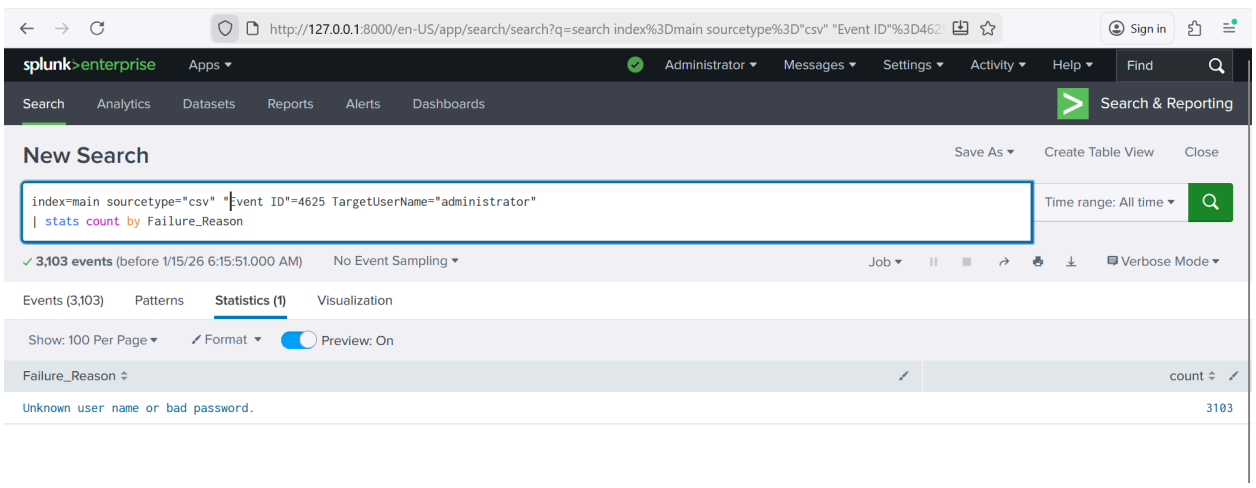

Figure 2 : SPL queries for Failure reasons

❖ **Additional Context**

- Source IP geolocation: Vietnam
- Event ID: 4625(Failed logon)
- Attack type: Credential based Brute-Force attack

5. Threat and Impact Assessment

- **Severity:** High
- **Impact:** No successful login observed, but credential exposure risk exists
- **Likelihood:** High - repeated attempts over a short time.

Note: Automated attacker with multiple source ports, external IP.

6. Recommended Actions

- Block source IP at firewall and endpoint security
- Implement lockout policy after a few failed attempts on the Administrator account.
- Monitor the targeted account for suspicious activity
- Escalate incident to Tier 2 SOC for containment or monitoring.

7. Tools and Method

- Splunk Enterprise : Log ingestion, Field extraction, SPL queries.
- Windows Event Log (CSV) **:** Source of events.
- Threat Intelligent: IP geolocation and reputation lookup